

Trade Tariffs and Exchange Rates in a Multilateral World

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September 17, 2026

1 Introduction

In the standard two-country setting, a tariff that improves the tariffing country's trade balance at unchanged exchange rates appreciates its currency under the Marshall–Lerner condition. With more than two countries, a tariff on one partner also reallocates expenditure across foreign suppliers, so bilateral exchange rates need not all move in the same direction. Such third-country effects are studied in the customs-union and preferential-tariff literature on the terms of trade (Viner, 1950; Mundell, 1964; Kemp and Wan, 1976). We show how these effects shape the exchange-rate consequences of tariffs in a multilateral setting.

In an N -country model, we show that the log-linearized equilibrium response can be written as $(-J)^{-1}F$, where F gives the trade-balance effects of a tariff at unchanged exchange rates and J is the Jacobian of trade balances with respect to exchange rates. Under a multilateral Marshall–Lerner (MML) condition on J , this separates the tariff disturbance from the exchange-rate adjustment required to restore equilibrium.

Under partner symmetry, we show that the response to any tariff vector further decomposes into average protection and relative treatment. Section 2 derives this decomposition without imposing a specific model of preferences or trade: average protection determines the tariffing country's nominal effective exchange rate (NEER), while relative treatment determines differences across bilateral rates. Section 3 then derives these responses in a nested-CES model. For a unilateral tariff, we show that the tariffing country appreciates against the target and its NEER appreciates, while its rate against an untariffed bystander can move in either direction. A bystander depreciation occurs only when substitution across foreign origins is sufficiently strong relative to the other effects of the tariff (Section 3.5). Section 4 extends the analysis to retaliatory and preferential tariffs. Proofs are in Appendix F.

2 Tariffs and Exchange-Rate Adjustment

This section does not rely on the nested-CES model of Section 3. It begins from a general differentiable trade system and adds restrictions where needed.

2.1 The bilateral case

Consider first two countries, A and B . Let e_{AB} be units of A 's currency per unit of B 's, so that a rise in e_{AB} is a depreciation of A , and write $\nu \equiv d \log e_{AB}$. Let A 's trade balance TB_A be a differentiable function of ν and of a tariff τ that A levies on B . Linearizing at a balanced free-trade baseline,

$$dT B_A = T B_{A\nu} \nu + T B_{A\tau} d\tau, \quad T B_{A\nu} \equiv \frac{\partial d T B_A}{\partial \nu}, \quad T B_{A\tau} \equiv \frac{\partial d T B_A}{\partial \tau} \Big|_{\nu \text{ fixed}}. \quad (1)$$

Suppose the tariff improves A 's trade balance at unchanged exchange rates, $T B_{A\tau} > 0$. Restoring balance then requires

$$\nu^* = -\frac{T B_{A\tau}}{T B_{A\nu}} d\tau, \quad (2)$$

which is an appreciation if and only if $T B_{A\nu} > 0$, that is, if and only if a depreciation improves A 's trade balance. This is the ordinary Marshall–Lerner condition; Section 3.2 shows that in the nested-CES model it takes the familiar elasticity form $\varepsilon_X + \varepsilon_M > 1$.

The two steps in (1)–(2) are the whole of the adjustment logic: the tariff first displaces the trade balance at unchanged exchange rates, and the exchange rate then moves to absorb that displacement. Figure 1 displays them.

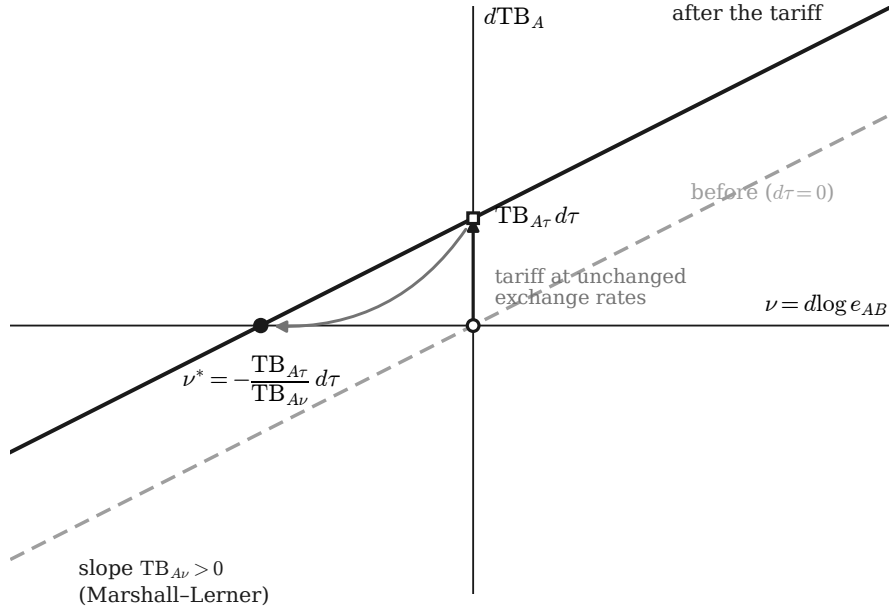


Figure 1: Bilateral equilibrium adjustment. The schedule is (1); its slope $T B_{A\nu}$ is positive exactly under the Marshall–Lerner condition. At unchanged exchange rates the tariff shifts the schedule up by $T B_{A\tau} d\tau > 0$ (open square). The currency then appreciates along the new schedule until balance is restored at $\nu^* = -(T B_{A\tau} / T B_{A\nu}) d\tau$.

2.2 The local disturbance–adjustment representation

Consider any differentiable multilateral trade environment with $N - 1$ independent trade-balance conditions, $TB_i = 0$ for $i \neq A$, which are each functions of exchange rates and tariffs. Stack the exchange-rate changes as $\nu \equiv (\nu_j)_{j \neq A}$, $\nu_j \equiv d \log e_{Aj}$, and the bilateral tariff changes as $d\tau \equiv (dt_{jk})_{j \neq k}$, so that ν and $d\tau$ are the perturbation vectors of the levels e_{Aj} and τ_{jk} . The conditions for $i \neq A$ are

$$J\nu + Gd\tau = 0, \quad J_{il} \equiv \frac{\partial dTB_i}{\partial \nu_l}, \quad G_{i,(jk)} \equiv \frac{\partial dTB_i}{\partial t_{jk}} \Big|_{\nu \text{ fixed}}. \quad (3)$$

A tariff experiment fixes a *pattern* of tariff changes a and a scalar *magnitude* ds , so that $d\tau = a ds$. Defining

$$F \equiv G a, \quad (4)$$

the vector of fixed-exchange-rate trade-balance disturbances generated by one unit of that pattern, the equilibrium conditions become

$$J\nu = -F ds, \quad \nu = (-J)^{-1} F ds. \quad (5)$$

Thus F , and more generally G , gives the trade-balance effects of the tariff at unchanged exchange rates, including substitution and income effects; J describes how trade balances respond to exchange-rate changes; and $(-J)^{-1}$ maps the disturbance into the equilibrium exchange-rate changes that absorb it.

At $N = 2$ there is a single condition, $TB_B = 0$. Since $dTB_B = -dTB_A$, the scalars in (3) are $-J = TB_{A\nu}$ and $F = -TB_{A\tau}$, and with $a = 1$ and $ds = d\tau$, (5) is exactly (2). The N -country system is the matrix analogue of that scalar problem.

2.3 The multilateral Marshall–Lerner condition

At $N = 2$, the requirement that exchange-rate adjustment stabilizes the trade balance is the scalar condition $-J = TB_{A\nu} > 0$ of Section 2.1. The N -country generalization requires that the adjustment mapping $(-J)^{-1}$ be well defined and nonnegative.

Definition 2.1 (Multilateral Marshall–Lerner condition). *The adjustment mechanism satisfies the multilateral Marshall–Lerner condition (MML) if*

$$-J \text{ is a nonsingular } M\text{-matrix:} \quad J_{il} \geq 0 \quad (l \neq i) \quad \text{and} \quad (-J)^{-1} \text{ exists with } (-J)^{-1} \geq 0. \quad (6)$$

By (5), under MML, each exchange-rate response is a nonnegative-weighted combination of the trade-balance effects at fixed exchange rates. Writing $\vartheta_{ij} \equiv [(-J)^{-1}]_{ij} \geq 0$,

$$\frac{d\nu_i}{ds} = \sum_j \vartheta_{ij} F_j, \quad (7)$$

so ϑ_{ij} measures the response of currency i to a unit trade-balance disturbance in country j . A uniformly signed F therefore cannot generate exchange-rate responses of the opposite sign. When F has mixed signs, the response of any individual exchange rate depends on the full vector of trade-balance effects rather than on the corresponding element F_i alone. The condition also implies $\det(-J) > 0$, which is used in Appendix D. At $N = 2$, the system is scalar and the condition reduces to $\text{TB}_{A\nu} > 0$, the ordinary Marshall–Lerner condition.

A sufficient condition for MML can be stated on the full Jacobian. Let ν_i , $i = 1, \dots, N$, denote the log change in the value of currency i in a common unit of account, so that $d \log e_{ij} = \nu_j - \nu_i$, and let $\tilde{J}_{il} \equiv \partial d\text{TB}_i / \partial \nu_l$ be the full $N \times N$ Jacobian. Normalizing $\nu_A = 0$ recovers the coordinates of Section 2.2, and J is $\tilde{J}$ with row and column A deleted. Each trade balance is homogeneous of degree one in currency values, and $\sum_i \text{TB}_i \equiv 0$ by Walras’ law, so at a balanced baseline

$$\sum_l \tilde{J}_{il} = 0 \quad \forall i, \quad \sum_i \tilde{J}_{il} = 0 \quad \forall l. \quad (8)$$

The row identity pins down each diagonal entry of $\tilde{J}$ from the off-diagonal entries, so a sign restriction on the latter suffices.

Assumption 1 (Gross substitutability of trade balances). $\tilde{J}_{il} \geq 0$ for all $l \neq i$.

An appreciation of currency l raises the price of l ’s goods; Assumption 1 requires that it weakly improve every other country’s trade balance once the associated income and expenditure effects are included. It is a restriction on trade-balance cross effects, not a condition on compensated demand alone. Together with (8), it makes J weakly diagonally dominant.

Lemma 2.2 (Sufficiency of gross substitutability). *Under Assumption 1, with J irreducible and $\tilde{J}_{iA} > 0$ for some i , MML (6) holds.*

The proof is in Appendix F.

2.4 Symmetric partners

Restrict attention to tariffs imposed by A , and write $t = (t_j)_{j \neq A}$ with $t_j \equiv dt_{Aj}$ for A ’s own tariff vector, so that G in (3) is $(N - 1) \times (N - 1)$. Suppose that A ’s trading partners are symmetric: relabeling them leaves the reduced system unchanged, $PJP' = J$ and $PGP' = G$ for every permutation matrix P of the $N - 1$ partners. This holds when the partners have identical sizes, trade weights, and parameters at the baseline; A itself need not be symmetric with them. Take $N \geq 3$.

Under partner symmetry, the tariff vector can affect exchange rates through only two components: average protection, the average tariff across partners, and relative treatment, the differences in tariffs across partners. Decompose

$$t = \bar{t} \mathbf{1} + \tilde{t}, \quad \bar{t} \equiv \frac{1}{N-1} \mathbf{1}' t, \quad \mathbf{1}' \tilde{t} = 0. \quad (9)$$

Componentwise this is

$$t_j = \bar{t} + (t_j - \bar{t}), \quad \tilde{t}_j \equiv t_j - \bar{t}. \quad (10)$$

We refer to $\bar{t}\mathbf{1}$ as *average protection* and to $\tilde{t}$ as *relative treatment*: average protection is the tariff level A applies to its partners on average, and relative treatment records how each partner is taxed relative to that average.

Let $d\nu_A^E \equiv \frac{1}{N-1} \sum_{j \neq A} \nu_j$ denote the equally weighted average of A 's bilateral responses, which under symmetry is A 's nominal effective exchange rate (20).

Proposition 2.3 (Average protection and relative treatment). *Suppose A 's partners are symmetric and $N \geq 3$. Then $-J$ and G each act as a scalar on $\text{span}\{\mathbf{1}\}$ and as a scalar on the zero-sum subspace $\{v : \mathbf{1}'v = 0\}$, so each has at most two distinct eigenvalues; write these scalars j_S, j_D and g_S, g_D respectively. If $j_S, j_D \neq 0$, the response to any tariff vector t imposed by A is*

$$\nu = \lambda_S \bar{t} \mathbf{1} + \lambda_D \tilde{t}, \quad \lambda_S \equiv \frac{g_S}{j_S}, \quad \lambda_D \equiv \frac{g_D}{j_D}, \quad (11)$$

so that average protection alone determines the effective exchange rate and relative treatment alone determines the dispersion of bilateral rates around it and the cross rates among partners:

$$d\nu_A^E = \lambda_S \bar{t}, \quad \nu_j - d\nu_A^E = \lambda_D \tilde{t}_j, \quad d \log e_{jk} = \lambda_D (t_k - t_j). \quad (12)$$

MML (6) holds if and only if

$$j_S > 0 \quad \text{and} \quad j_D \geq j_S. \quad (13)$$

The reason the response takes this two-term form is that any matrix invariant to relabeling the partners has the form $aI + b\mathbf{1}\mathbf{1}'$. With $\mathbf{1}'\mathbf{1} = N - 1$ and $\mathbf{1}'\tilde{t} = 0$,

$$(aI + b\mathbf{1}\mathbf{1}')t = (a + (N - 1)b) \bar{t} \mathbf{1} + a \tilde{t}.$$

The matrix therefore acts separately on average protection and relative treatment, with eigenvalues $a + (N - 1)b$ on $\text{span}\{\mathbf{1}\}$ and a on the zero-sum subspace. For $-J$ and G these are

$$\begin{aligned} -J = a_J I + b_J \mathbf{1}\mathbf{1}' : \quad j_S &= a_J + (N - 1)b_J, \quad j_D = a_J, \\ G = a_G I + b_G \mathbf{1}\mathbf{1}' : \quad g_S &= a_G + (N - 1)b_G, \quad g_D = a_G. \end{aligned} \quad (14)$$

Since relative treatment sums to zero, it also has no effect on the equally weighted average of the bilateral exchange rates, which is the first equality in (12).

Proposition 2.3 is model-free: it says only that exchangeability reduces the adjustment problem to two scalars, without saying what those scalars are. Section 3 computes them for the nested-CES model.

3 The Nested-CES Model

We now specify the N -country model whose matrices J and G the rest of the paper characterizes, and specialize the general results of Section 2 to it.

3.1 Production, preferences, and equilibrium

Countries are indexed by $i, j \in \{1, \dots, N\}$. Where useful, we use A for the tariffing country, B for the tariff target, and C for an untariffed bystander. Each country produces a country-specific tradable and a non-tradable from labor:

$$Y_{T_i} = A_{T_i} L_{T_i}, \quad Y_{N_i} = A_{N_i} L_{N_i}, \quad L_{T_i} + L_{N_i} = L_i,$$

with perfect competition and the producer-price normalization $P_{T_i} = 1$ in i 's currency, so

$$w_i = A_{T_i}, \quad P_{N_i} = A_{T_i}/A_{N_i}.$$

Linear production, perfect competition, and the producer-price normalization fix wages in own currency, so relative international prices adjust through exchange rates.

Preferences are nested CES:

$$\begin{aligned} C_i &= C_{T_i}^{\alpha_T} C_{N_i}^{1-\alpha_T} && \text{(tradable share } \alpha_T), \\ C_{T_i} &= \left[\alpha_D^{1/\eta} C_{T_i i}^{\frac{\eta-1}{\eta}} + (1-\alpha_D)^{1/\eta} M_i^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}} && \text{(home weight } \alpha_D, \text{ elasticity } \eta), \\ M_i &= \left[\sum_{j \neq i} \beta_{ij}^{1/\rho} C_{T_j i}^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1}} && \text{(origin weights } \beta_{ij}, \text{ elasticity } \rho). \end{aligned}$$

The nesting is chosen precisely to separate the two substitution margins that the adjustment problem turns on: the elasticity η governs substitution between domestic and imported tradables, while ρ governs substitution across foreign origins conditional on importing. The latter margin is inactive at $N = 2$ and active at $N \geq 3$.

The *symmetric benchmark* imposes $\beta_{ij} = 1/(N-1)$, common parameters $(\alpha_T, \alpha_D, \eta, \rho)$, and equal endowments and productivities. Home bias remains through α_D . Full symmetry between domestic and foreign tradables would additionally require $\rho = \eta$ and $\alpha_D = 1/N$, which we do not impose.

Define e_{ij} as units of i 's currency per unit of j 's, with $e_{ii} = 1$ and $e_{ij} = e_{ik}e_{kj}$. A rise in e_{ij} is therefore a depreciation of i against j . Log changes in bilateral exchange rates relative to A are denoted by $\nu_j \equiv d \log e_{Aj}$, with $\nu_A = 0$.

Tariffs $\tau_{ij} \geq 0$ are ad valorem and are levied by i on imports from j , with $\tau_{ii} = 0$. Consumer prices in i 's currency are $p_{ij} = (1+\tau_{ij})e_{ij}$ and $p_{ii} = 1$, where the latter follows from the normalization of the domestic tradable price. The corresponding price indices are

$$P_{M_i} = \left[\sum_{j \neq i} \beta_{ij} p_{ij}^{1-\rho} \right]^{\frac{1}{1-\rho}}, \quad P_{T_i} = \left[\alpha_D p_{ii}^{1-\eta} + (1-\alpha_D) P_{M_i}^{1-\eta} \right]^{\frac{1}{1-\eta}}, \quad P_i = \kappa_i P_{T_i}^{\alpha_T} P_{N_i}^{1-\alpha_T}. \quad (15)$$

Expenditure shares of income are

$$s_{ii} = \alpha_T \alpha_D \left(\frac{p_{ii}}{P_{T_i}} \right)^{1-\eta}, \quad s_{ij} = \alpha_T (1-\alpha_D) \left(\frac{P_{M_i}}{P_{T_i}} \right)^{1-\eta} \beta_{ij} \left(\frac{p_{ij}}{P_{M_i}} \right)^{1-\rho}, \quad j \neq i, \quad (16)$$

so $s_{ij}I_i$ is tariff-inclusive expenditure at consumer prices and $s_{ij}I_i/(1 + \tau_{ij})$ is the tariff-exclusive border value. Tariff revenue is rebated lump-sum:

$$I_i = w_i L_i + \sum_{j \neq i} \frac{\tau_{ij}}{1 + \tau_{ij}} s_{ij} I_i \implies I_i = \frac{w_i L_i}{1 - \sum_{j \neq i} \frac{\tau_{ij}}{1 + \tau_{ij}} s_{ij}}. \quad (17)$$

Equation (17) determines income in i 's own currency. To aggregate bilateral flows across countries, take A 's currency as the numeraire and define common-currency income as $\tilde{I}_i \equiv e_{Ai} I_i$. Expenditure shares are unchanged by this conversion. Since tariff revenue is a domestic transfer, trade balances are valued at tariff-exclusive border prices in the common currency:

$$\text{TB}_i = \sum_{k \neq i} f_{ki} - \sum_{j \neq i} f_{ij}, \quad f_{ij} \equiv \frac{s_{ij} \tilde{I}_i}{1 + \tau_{ij}}, \quad (18)$$

where f_{ij} is the border value, in A 's currency, of i 's imports from j .

By Walras' law, only $N - 1$ trade-balance conditions are independent. We assume no internationally traded assets, so $\text{TB}_i = 0$ for all i . Given $\{\tau_{ij}\}$, the $N - 1$ independent exchange rates therefore solve $\text{TB}_i = 0$ for $i \neq A$.

Terms of trade and real exchange rates are

$$\text{ToT}_{ij} \equiv \frac{P_{T_i}}{e_{ij} P_{T_j}} = \frac{1}{e_{ij}}, \quad q_{ij} \equiv \frac{e_{ij} P_j}{P_i}. \quad (19)$$

Nominal effective exchange-rate changes are

$$d\nu_i^E \equiv \sum_{j \neq i} w_{ij} d \log e_{ij}, \quad w_{ij} = \frac{f_{ij} + f_{ji}}{\sum_{k \neq i} (f_{ik} + f_{ki})}, \quad (20)$$

with $w_{ij} = 1/(N - 1)$ at the symmetric benchmark. Under the normalization of domestic tradable producer prices, $d \log e_{ij} = -d \log \text{ToT}_{ij}$, so the nominal exchange-rate results can equivalently be expressed in terms of the terms of trade.

3.2 The two-country case

Appendix B gives the general linearization, including the $N = 2$ calculation used here. Write $m \equiv \alpha_T(1 - \alpha_D)$ for the import share of income and $\nu_B \equiv d \log e_{AB}$, and suppose A imposes $d\tau = dt_{AB}$. Each country has a single import origin, so the cross-origin margin in (16) is inactive. Define

$$\rho_1^* \equiv 1 + \alpha_D(\eta - 1) - \alpha_T(1 - \alpha_D) = \alpha_D \eta + (1 - \alpha_D)(1 - \alpha_T) > 0. \quad (21)$$

Linearizing (18) at the balanced free-trade baseline, the equilibrium condition is

$$d\text{TB}_A = \underbrace{m D_2 \nu_B}_{\text{exchange-rate effect}} + \underbrace{m \rho_1^* d\tau}_{\text{tariff effect at fixed exchange rates}} = 0, \quad D_2 \equiv 2\alpha_D(\eta - 1) + 1, \quad (22)$$

and therefore

$$\boxed{\left. \frac{d \log e_{AB}}{d\tau} \right|_{\tau=0} = -\frac{\rho_1^*}{D_2}} \quad (23)$$

Comparing (22) with (1), the generic bilateral coefficients of Section 2.1 are $TB_{A\nu} = mD_2$ and $TB_{A\tau} = m\rho_1^*$, and (23) is (2). In the notation of (3), $-J = mD_2$ and $F = -m\rho_1^*$.

The coefficient ρ_1^* summarizes the trade-balance effect of the tariff at unchanged exchange rates. It reflects substitution between domestic and imported goods, governed by η , together with the associated income and expenditure effects. The second form in (21) separates the two: $\alpha_D\eta$ is switching toward the home tradable, and $(1 - \alpha_D)(1 - \alpha_T)$ reflects the part of tariff-induced income and expenditure that falls on nontradables rather than on imported tradables. The elasticity ρ , which governs substitution across foreign import origins, is inactive with only two countries. It becomes relevant for $N \geq 3$, when a tariff on one partner can redirect import demand toward other foreign suppliers.

At the symmetric two-country benchmark, the reduced-form export and import elasticities are $\varepsilon_X = \varepsilon_M = 1 + \alpha_D(\eta - 1)$, so that $D_2 = \varepsilon_X + \varepsilon_M - 1$. An appreciation therefore restores trade balance after a tariff if and only if the textbook Marshall–Lerner condition $\varepsilon_X + \varepsilon_M > 1$ holds. In our nested-CES model this condition is $D_2 > 0$, which holds whenever $\eta \geq 1$. So D_2 is the object that decides whether exchange-rate adjustment is stabilizing, and ρ_1^* the object that measures the disturbance it has to absorb. Appendix A gives the exact finite-tariff Cobb–Douglas solution.

3.3 The N -country characterization

At the symmetric benchmark A 's partners are symmetric in the sense of Section 2.4, so Proposition 2.3 applies and the entire adjustment problem reduces to the two pairs (j_S, j_D) and (g_S, g_D) . The following theorem evaluates them in the nested-CES model; Appendix B gives the linearized system from which they are obtained.

Theorem 3.1 (Tariffs and exchange rates at the symmetric benchmark). *Consider the nested-CES model of Section 3 at the symmetric benchmark, with $N \geq 3$, and let $t = (t_j)_{j \neq A}$ be any vector of tariffs imposed by A , decomposed as in (10). Define*

$$D_N \equiv N\alpha_D(\eta - 1) + (N - 2)\rho + 1. \quad (24)$$

Then MML (6) holds if and only if $D_N > 0$; under the maintained parameter restrictions, $\eta \geq 1$ is sufficient for $D_N > 0$. If $D_N > 0$, the coefficients of Proposition 2.3 are

$$\lambda_S = \frac{g_S}{j_S} = -\frac{(N - 1)\rho_1^*}{D_N}, \quad \lambda_D = \frac{g_D}{j_D} = -\frac{(N - 1)\rho}{ND_N}, \quad (25)$$

both strictly negative, and the equilibrium exchange-rate response to t is

$$\nu_j = \lambda_S \bar{t} + \lambda_D (t_j - \bar{t}), \quad (26)$$

with $d\nu_A^E = \lambda_S \bar{t}$, $\nu_j - d\nu_A^E = \lambda_D \tilde{t}_j$ and $d \log e_{jk} = \lambda_D (t_k - t_j)$ as in (12).

Thus D_N is the N -country counterpart of the two-country adjustment coefficient $D_2 = \varepsilon_X + \varepsilon_M - 1$. Average protection determines the effective exchange-rate response, while relative treatment determines deviations of bilateral rates from that common response. In particular, changes in relative treatment that leave $\bar{t}$ unchanged do not affect A 's effective exchange rate. Since $\lambda_D < 0$, $t_j > t_k$ implies $\nu_j < \nu_k$: A appreciates more against partners facing higher tariffs. Appendix B gives the exact condition for a depreciation against a given partner and the responses when A taxes several partners.

3.4 Adjustment coefficients and parameters

The coefficients in Theorem 3.1 can be understood by separating the tariff disturbance from exchange-rate adjustment. For the tariff disturbance,

$$g_S = -\frac{m\rho_1^*}{N-1}, \quad g_D = -\frac{m\rho}{N-1}, \quad (27)$$

and for the exchange-rate adjustment,

$$j_S = \frac{mD_N}{(N-1)^2}, \quad j_D = N \frac{mD_N}{(N-1)^2}; \quad (28)$$

Appendix B derives both. Their ratios are

$$\frac{g_D}{g_S} = \frac{\rho}{\rho_1^*}, \quad \frac{j_D}{j_S} = N. \quad (29)$$

The first is the disturbance ratio: the relative loading of the tariff on the relative-treatment and average-protection disturbances. It compares substitution across foreign origins, ρ , with the home-switching and spending effects collected in ρ_1^* , and it does not depend on N . The second is the adjustment ratio: the adjustment eigenvalue on the relative-treatment mode is N times that on the common mode, so a disturbance of a given modal size requires only $1/N$ as much exchange-rate adjustment. Combining the two ratios gives

$$\frac{\lambda_D}{\lambda_S} = \frac{g_D}{g_S} \cdot \frac{j_S}{j_D} = \frac{\rho}{N\rho_1^*}. \quad (30)$$

In the nested-CES benchmark, the factor N therefore comes from the adjustment ratio, not from dilution of the tariff disturbance across more partners: the disturbance ratio $g_D/g_S = \rho/\rho_1^*$ is the same at every N .¹

The parameters enter the coefficients at distinct margins. The elasticity η governs substitution between domestic and imported tradables and enters only through ρ_1^* and D_N ; it is the sole substitution margin at $N = 2$. The elasticity ρ governs substitution across foreign origins and enters *both* sides of (5): it raises the relative-treatment disturbance g_D and, through D_N , both adjustment coefficients. The weight α_T appears only in ρ_1^* , while α_D enters both ρ_1^* and D_N .

¹The factor N is derived here for the symmetric nested-CES benchmark. Appendix D considers asymmetric baselines.

3.5 A unilateral tariff

Suppose A raises its tariff on B alone, so that $t = d\tau \mathbf{e}_B$. Then

$$\bar{t} = \frac{d\tau}{N-1}, \quad \tilde{t}_B = \frac{N-2}{N-1}d\tau, \quad \tilde{t}_C = -\frac{d\tau}{N-1} \quad \text{for each bystander } C.$$

Substituting into (26), A appreciates against the target,

$$\frac{d \log e_{AB}}{d\tau} = -\frac{N\rho_1^* + (N-2)\rho}{ND_N} < 0, \quad (31)$$

while the responses against an untargeted bystander are

$$\frac{d \log e_{AC}}{d\tau} = \frac{\rho - N\rho_1^*}{ND_N}, \quad \frac{d \log e_{BC}}{d\tau} = \frac{(N-1)\rho}{ND_N} > 0. \quad (32)$$

Average protection alone carries the effective exchange rate, which at this benchmark appreciates for every $N \geq 2$ and every admissible ρ :

$$\frac{d\nu_A^E}{d\tau} = \frac{\nu_B + (N-2)\nu_C}{N-1} = -\frac{\rho_1^*}{D_N} < 0. \quad (33)$$

By (32), the bilateral bystander response reverses sign, so that A depreciates against C , if and only if

$$\boxed{\rho > \rho_N^* \equiv N\rho_1^*}. \quad (34)$$

The reversal cannot occur when the two substitution elasticities coincide and $\alpha_D \geq 1/N$. If $\rho = \eta = \sigma$ and $\alpha_D \geq 1/N$, then

$$N\rho_1^* - \sigma = \sigma(N\alpha_D - 1) + N(1 - \alpha_D)(1 - \alpha_T) \geq 0, \quad (35)$$

so $\rho \leq N\rho_1^*$.² Without imposing $\rho = \eta$, the same restriction gives $N\rho_1^* - \eta = \eta(N\alpha_D - 1) + N(1 - \alpha_D)(1 - \alpha_T) \geq 0$, so $\rho > \eta$ is necessary, but not sufficient, for the reversal. Appendix C gives the corresponding real exchange-rate results.

The bystander threshold differs from the condition under which the bystander's trade balance improves at unchanged exchange rates. From (49),

$$F_B = -\frac{m[\rho_1^* + (N-2)\rho]}{(N-1)^2} < 0, \quad F_C = \frac{m(\rho - \rho_1^*)}{(N-1)^2}. \quad (36)$$

so

$$F_C > 0 \iff \rho > \rho_1^*, \quad \text{whereas} \quad \nu_C > 0 \iff \rho > N\rho_1^*.$$

A favorable fixed-exchange-rate disturbance for the bystander is therefore not sufficient for a depreciation of A 's currency against it. The first threshold is the disturbance ratio $g_D/g_S = \rho/\rho_1^*$ crossing one and is independent of N ; the second is $\lambda_D/\lambda_S = \rho/(N\rho_1^*)$ crossing one and carries

²Equality in (35) requires $\alpha_D = 1/N$ and $\alpha_T = 1$. At $\alpha_D = 1/N$ with $\rho = \eta = \sigma$, $d \log e_{AC}/d\tau = -(1 - \alpha_T)/(N\sigma)$, which approaches zero from below as $\alpha_T \rightarrow 1$; see Appendix B.

the adjustment ratio $j_D/j_S = N$. Figure 2 shows the same comparison as a decomposition of the response vector.

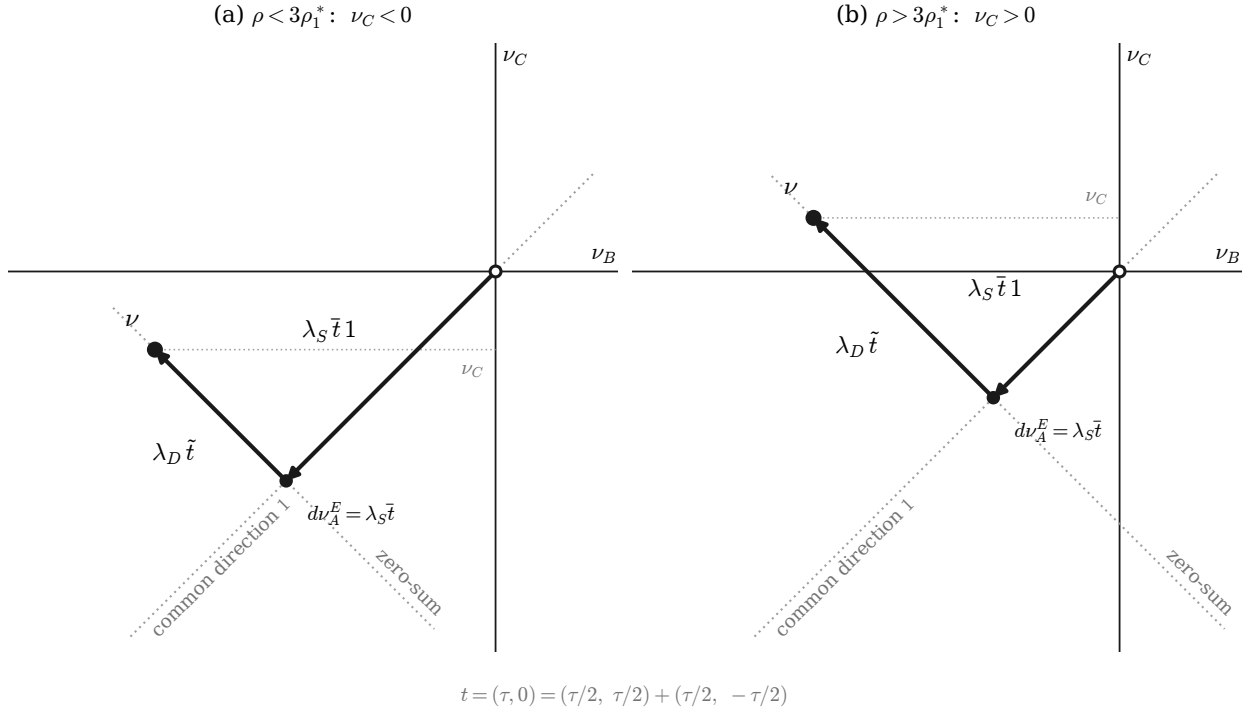


Figure 2: Average protection and relative treatment at $N = 3$. The unilateral tariff $t = (\tau, 0)$ is split as $(\tau/2, \tau/2) + (\tau/2, -\tau/2)$. The average-protection component moves both bilateral rates equally along the constant direction **1** and carries the whole effective-rate response $d\nu_A^E = \lambda_S \bar{t}$; the relative-treatment component moves them in opposite directions along the zero-sum direction and leaves the effective rate unchanged. The two components are orthogonal, so the average-protection point is the projection of ν onto **1**. Panel (a): $\rho < 3\rho_1^*$; panel (b): $\rho > 3\rho_1^*$. Drawn at $\alpha_D = 0.7$, $\alpha_T = 0.6$, $\eta = 1.5$, with $\rho = 2.2$ in panel (a) and $\rho = 5$ in panel (b), so that $\rho_1^* = 1.17$ and $3\rho_1^* = 3.51$.

4 Trade Wars and Preferential Tariffs

Because the analysis is local, the effects of simultaneous tariff changes by several countries can be computed separately and superposed to first order. At the symmetric benchmark, each tariffing country's contribution follows from Theorem 3.1.

4.1 Trade wars

Bilateral retaliation. Suppose A raises its tariff on B by $d\tau$ and B raises its tariff on A by the same amount. The two effects on the A – B rate cancel, while for any bystander C ,

$$\left. \frac{d \log e_{AB}}{d\tau} \right|^w = 0, \quad \left. \frac{d \log e_{AC}}{d\tau} \right|^w = \frac{\rho - \rho_1^*}{D_N}, \quad \left. \frac{d\nu_A^E}{d\tau} \right|^w = \frac{N-2}{N-1} \frac{\rho - \rho_1^*}{D_N}. \quad (37)$$

The A – B rate is therefore unchanged, but each belligerent’s effective exchange rate can still move through its rates against the bystanders. Relative to a unilateral tariff, retaliation lowers the bystander threshold from $N\rho_1^*$ to ρ_1^* : B ’s tariff on A contributes an additional relative-treatment effect on the A – C rate, which removes the factor N from the threshold.

Global retaliation. Suppose instead that A raises its tariff by $d\tau$ on every partner and each partner raises its tariff by $d\tau$ on A .

Proposition 4.1 (Trade wars). *At the symmetric benchmark of the nested-CES model, with $N \geq 3$ and under MML:*

(i) (Bilateral retaliation.) *In a symmetric bilateral trade war between A and B , the belligerents’ effective exchange rates depreciate, and every bystander appreciates against both, if and only if $\rho > \rho_1^*$. The threshold is independent of N .*

(ii) (Global retaliation.) *If A taxes every partner and every partner taxes A , each at rate $d\tau$, then*

$$\frac{d \log e_{Aj}}{d\tau} = \frac{d\nu_A^E}{d\tau} = \frac{(N-2)(\rho - \rho_1^*)}{D_N} \quad \forall j, \quad (38)$$

while all exchange rates among A ’s partners are unchanged. Hence A depreciates against every partner if and only if $\rho > \rho_1^$.*

The threshold in (ii) is the same as under bilateral retaliation.

4.2 Preferential tariffs and competing exporters

A standard benchmark in the preferential-trade-agreement literature has one importing country and two competing exporters, following Viner (1950) and Mundell (1964). Importer A is endowed only with the numeraire good y , while exporters B and C are endowed with good x . Preferences are quasilinear, $U_i = u_i(x_i) + y_i$. Country A levies specific tariffs t_B and t_C . Let p be the domestic price of x in A , and let q_j be exporter j ’s price of x in terms of the numeraire, which is j ’s terms of trade. Market clearing requires

$$p = q_B + t_B = q_C + t_C, \quad M_A(p) = E_B(p - t_B) + E_C(p - t_C), \quad M'_A < 0, \quad E'_j > 0, \quad (39)$$

where $M_A(p) = x_A^d(p)$ and $E_j(q_j) = \omega_j - x_j^d(q_j)$. Differentiating with respect to t_B gives

$$\frac{dp}{dt_B} = \frac{E'_B}{E'_B + E'_C - M'_A} \in (0, 1), \quad \frac{dq_B}{dt_B} = -\frac{E'_C - M'_A}{E'_B + E'_C - M'_A} < 0, \quad \frac{dq_C}{dt_B} = \frac{dp}{dt_B} > 0. \quad (40)$$

so a tariff on imports from B worsens B ’s terms of trade and improves C ’s.

In our model an improvement in C ’s terms of trade corresponds to an increase in e_{AC} by (19). In the competing-exporter model this occurs unconditionally after a tariff on B , whereas at the symmetric benchmark of the nested-CES model (34) requires $\rho > N\rho_1^*$. A preferential tariff

reduction gives the corresponding result with the signs reversed: if A lowers its tariff on B alone, $t = -\tau \mathbf{e}_B$, a non-member C 's terms of trade worsen if and only if $\rho > N\rho_1^*$.

The difference reflects the economic environment. In the competing-exporter model, A does not produce the importable good, so a tariff reallocates demand between the two foreign suppliers without inducing substitution toward a domestically produced import-competing good; the model isolates substitution across foreign suppliers. In our model, substitution across foreign import origins, governed by ρ , operates alongside substitution toward the domestic import-competing tradable, governed by η , and the associated income effects, both collected in ρ_1^* . The third-country terms-of-trade effect has the competing-exporter sign only when the first margin is strong relative to the second. The threshold rises with the domestic-import substitution term $\alpha_D(\eta - 1)$ and falls with the import share $\alpha_T(1 - \alpha_D)$; at $N = 3$ it is $3\rho_1^*$.

The two environments are not formally nested. The competing-exporter model has a common importable supplied by B and C and quasilinear preferences, while our model has country-specific differentiated tradables and nested-CES preferences, so the comparison is one of mechanisms. Competing-exporter models such as Bagwell and Staiger (1999) and Ornelas (2005) also abstract from trade between the foreign suppliers and use quasilinear preferences, while bloc and endowment models such as Krugman (1991) and Bond and Syropoulos (1996) abstract from domestic production of the importable.

5 Conclusion

A tariff affects exchange rates through two steps. At unchanged exchange rates it changes countries' trade balances, summarized by F ; exchange rates then adjust through $(-J)^{-1}$. In the two-country case this reduces to the familiar Marshall–Lerner result: when a tariff improves the trade balance at unchanged exchange rates, the elasticity condition ensures that the tariffing country's currency appreciates.

With three or more countries, a tariff can also redirect expenditure across foreign suppliers. Under symmetry, the response to any tariff vector separates into an average-protection component and a relative-treatment component. Average protection determines the effective exchange-rate response, while relative treatment determines deviations of bilateral rates from that common response. At the symmetric benchmark of the nested-CES model, the tariffing country continues to appreciate against the targeted partner, but its exchange rate against an untariffed partner can move in either direction.

In that benchmark, a unilateral tariff on one partner always appreciates the tariffing country's effective exchange rate; under asymmetric trade weights this need not hold (Appendix D). Its bilateral rate against an untariffed bystander reverses only if

$$\rho > N\rho_1^*,$$

whereas the bystander's trade balance at unchanged exchange rates improves already at $\rho > \rho_1^*$. The factor N comes from multilateral exchange-rate adjustment, $j_D/j_S = N$, rather than from dilution of the initial tariff disturbance, whose ratio $g_D/g_S = \rho/\rho_1^*$ does not depend on N .

Other tariff configurations alter this comparison. Under symmetric bilateral retaliation, the bystander threshold falls to ρ_1^* and is independent of N . In the standard competing-exporter benchmark, substitution across foreign suppliers generates an unambiguous third-country terms-of-trade effect; in our model, that effect interacts with domestic-import substitution and income effects and therefore need not have an unconditional sign.

The main implication is that the familiar two-country exchange-rate result survives multilateralization most cleanly at the level of the effective exchange rate. Bilateral exchange rates also reflect how trade policy differs across partners, and their responses need not inherit the sign of the aggregate adjustment.

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A Exact Cobb–Douglas solution ($N = 2$)

At $\eta = 1$, expenditure shares are fixed at their weights. From (17), $I_A = w_A L_A / [1 - \frac{\tau}{1+\tau} m]$, and $TB_A = 0$ becomes $e m w_B L_B = m I_A / (1 + \tau)$:

$$e(\tau) = \frac{w_A L_A}{w_B L_B} \cdot \frac{1}{1 + (1 - m)\tau}, \quad \left. \frac{d \log e}{d\tau} \right|_{\tau=0} = -(1 - m) = -\left. \frac{\rho_1^*}{D_2} \right|_{\eta=1}. \quad (41)$$

Thus $e(\tau)$ is decreasing in τ , so the sign of the linearized response in (23) is exact for all tariff levels in this case. Here $\eta = 1$ fixes the elasticity, not the weight α_D .

B The Linearized System

At the balanced free-trade baseline, let $m \equiv \alpha_T(1 - \alpha_D)$, $b_{ij} \equiv \beta_{ij}$, and $f_{ij} = s_{ij}\tilde{I}_i$, with $\sum_k f_{ki} = \sum_j f_{ij}$. All incomes and flows are measured in the common currency. Linearizing the demand, income, and trade-balance equations gives

$$dp_{ij} = \nu_j - \nu_i + dt_{ij} \quad (dp_{ii} = 0), \quad (42)$$

$$d \log P_{M_i} = \sum_{j \neq i} b_{ij} dp_{ij}, \quad d \log P_{T_i} = (1 - \alpha_D) d \log P_{M_i}, \quad (43)$$

$$d \log s_{ii} = -(1 - \eta)(1 - \alpha_D) d \log P_{M_i}, \quad (44)$$

$$d \log s_{ij} = (1 - \rho)(dp_{ij} - d \log P_{M_i}) + (1 - \eta)\alpha_D d \log P_{M_i}, \quad (45)$$

$$d \log \tilde{I}_i = \nu_i + \sum_{j \neq i} s_{ij} dt_{ij}, \quad (46)$$

$$d \text{TB}_i = \sum_{k \neq i} f_{ki}(d \log s_{ki} + d \log \tilde{I}_k - dt_{ki}) - \sum_{j \neq i} f_{ij}(d \log s_{ij} + d \log \tilde{I}_i - dt_{ij}). \quad (47)$$

These equations determine the model-specific matrices J and G in (3). At $N = 2$ they reduce to the calculation in Section 3.2.

$$\tilde{J}_{il} = \frac{m}{(N-1)^2} D_N \quad (l \neq i), \quad D_N \equiv N\alpha_D(\eta - 1) + (N-2)\rho + 1. \quad (48)$$

Lemma B.1 below gives the derivation.

Lemma B.1 (Full Jacobian at the symmetric baseline). *At the symmetric balanced baseline of Appendix B, where $b_{ij} = 1/(N-1)$ and $f_{ij} = m/(N-1)$, the full $N \times N$ Jacobian of the nested-CES model is*

$$\tilde{J}_{il} = \frac{m}{(N-1)^2} D_N \quad (l \neq i), \quad \tilde{J}_{ii} = -\frac{m}{N-1} D_N, \quad D_N = N\alpha_D(\eta - 1) + (N-2)\rho + 1.$$

Proof. Set $dt_{jk} = 0$, so $dp_{ij} = \nu_j - \nu_i$ and $d \log \tilde{I}_i = \nu_i$. Writing $S \equiv \sum_j \nu_j$, (43) gives

$$d \log P_{M_k} = \frac{S - N\nu_k}{N-1}, \quad \frac{\partial d \log P_{M_k}}{\partial \nu_l} = \frac{1 - N\delta_{kl}}{N-1}.$$

On the import side,

$$\sum_{j \neq i} (dp_{ij} - d \log P_{M_i}) = 0,$$

so the $(1 - \rho)$ terms cancel when (45) is summed over $j \neq i$. Thus,

$$\sum_{j \neq i} d \log s_{ij} = (N-1)(1 - \eta)\alpha_D d \log P_{M_i}.$$

Substituting into (47) gives

$$dTB_i = \frac{m}{N-1} \left[\sum_{k \neq i} (d \log s_{ki} + \nu_k) - (N-1)(1-\eta)\alpha_D d \log P_{M_i} - (N-1)\nu_i \right].$$

Fix $l \neq i$ and differentiate term by term:

$$\frac{\partial}{\partial \nu_l} \sum_{k \neq i} d \log s_{ki} = -(1-\rho) \frac{N-2}{N-1} - \frac{(1-\eta)\alpha_D}{N-1}.$$

The export income terms contribute one, the import-side price-index term contributes $-(1-\eta)\alpha_D$, and the import income term contributes zero. Collecting terms gives

$$\tilde{J}_{il} = \frac{m}{(N-1)^2} \left[(N-1) - (1-\rho)(N-2) + N\alpha_D(\eta-1) \right] = \frac{m}{(N-1)^2} D_N.$$

The diagonal follows from (8):

$$\tilde{J}_{ii} = -(N-1)\tilde{J}_{il} = -\frac{m}{N-1} D_N.$$

At $N = 2$ this gives $\tilde{J}_{AB} = mD_2$, the coefficient in (22); deleting row and column A at $N = 3$ reproduces J in (54). $\square$

The two-country case. At the balanced free-trade baseline with normalized incomes, $s_{AB} = s_{BA} = m$. At $N = 2$ each country has a single import origin, so $dp_{AB} = d \log P_{M_A} = \nu_B + d\tau$ and $dp_{BA} = d \log P_{M_B} = -\nu_B$. Log-differentiating (16) and (17), with incomes in the common currency,

$$\begin{aligned} d \log s_{AB} &= (1-\eta)\alpha_D(\nu_B + d\tau), & d \log s_{BA} &= -(1-\eta)\alpha_D \nu_B, \\ d \log \tilde{I}_A &= m d\tau, & d \log \tilde{I}_B &= \nu_B. \end{aligned}$$

Substituting into (18) and using that the tariff wedge lowers the border value of A 's imports by $d\tau$,

$$dTB_A = m \left[(1 + 2\alpha_D(\eta-1))\nu_B + (1 - m + \alpha_D(\eta-1))d\tau \right],$$

which is (22), with $D_2 = 1 + 2\alpha_D(\eta-1)$ and $\rho_1^* = 1 - m + \alpha_D(\eta-1)$, and (23) follows. For the elasticity form, hold income fixed and let $M(p)$ denote import volume at consumer price $p = p_{AB}$. Since import value is $s_{AB}I_A$,

$$\varepsilon_M \equiv -\frac{\partial \log M}{\partial \log p} = 1 - \frac{\partial \log s_{AB}}{\partial \log p} = 1 - (1-\eta)\alpha_D = 1 + \alpha_D(\eta-1).$$

By symmetry, $\varepsilon_X = 1 + \alpha_D(\eta-1)$. Differentiating $TB_A(e, \tau) = X(1/e) - eM((1+\tau)e)$ at $\tau = 0$ and using balanced trade, $X = eM$, gives

$$\frac{\partial TB_A}{\partial e} = M(\varepsilon_X + \varepsilon_M - 1).$$

Hence $\varepsilon_X + \varepsilon_M - 1 = 2\alpha_D(\eta - 1) + 1 = D_2$, so the stability condition $D_2 > 0$ is exactly $\varepsilon_X + \varepsilon_M > 1$. Since $\rho_1^* > 0$ by (21), the slope $-\rho_1^*/D_2$ is negative whenever $D_2 > 0$.

The tariff disturbance and the four eigenvalues. At unchanged exchange rates, the effect on partner i 's trade balance follows from (47). Since only i 's exports to A are directly affected,

$$d\text{TB}_i|_{\nu \text{ fixed}} = f_{Ai}(d\log s_{Ai} + d\log \tilde{I}_A - t_i) = \frac{m}{N-1} \left[-\rho_1^* \bar{t} - \rho(t_i - \bar{t}) \right]. \quad (49)$$

Reading off the coefficients on $\bar{t}$ and $\tilde{t}_i$ gives the tariff-effect eigenvalues (27). Deleting row and column A from (48) gives the exchange-rate adjustment matrix

$$-J = c(NI - \mathbf{1}\mathbf{1}'), \quad c \equiv \frac{mD_N}{(N-1)^2}, \quad (50)$$

whose eigenvalues are $j_S = c$ on $\text{span}\{\mathbf{1}\}$ and $j_D = Nc$ on the zero-sum subspace, which is (28). Dividing (27) by (28) gives (25).

The unilateral tariff. For $t = d\tau \mathbf{e}_B$, $\bar{t} = d\tau/(N-1)$, $\tilde{t}_B = \frac{N-2}{N-1}d\tau$ and $\tilde{t}_C = -\frac{d\tau}{N-1}$. By Theorem 3.1,

$$\nu_B = \frac{\lambda_S + (N-2)\lambda_D}{N-1} d\tau = -\frac{N\rho_1^* + (N-2)\rho}{ND_N} d\tau, \quad \nu_C = \frac{\lambda_S - \lambda_D}{N-1} d\tau = \frac{\rho - N\rho_1^*}{ND_N} d\tau,$$

while $d\log e_{BC} = \nu_C - \nu_B = -\lambda_D d\tau$ and $d\nu_A^E = \lambda_S d\tau/(N-1) = -(\rho_1^*/D_N) d\tau$. Since $\rho_1^* > 0$ by (21) and $D_N > 0$ under MML, these give (31)–(34).

Derivation of (35). With $\rho = \eta = \sigma$,

$$N\rho_1^* = N[1 + \alpha_D(\sigma - 1) - \alpha_T(1 - \alpha_D)],$$

so

$$N\rho_1^* - \sigma = \sigma(N\alpha_D - 1) + N(1 - \alpha_D)(1 - \alpha_T).$$

For $\alpha_D \geq 1/N$ both terms are nonnegative, with both zero only at $\alpha_D = 1/N$, $\alpha_T = 1$; hence $\rho = \sigma \leq N\rho_1^*$. Since ρ_1^* does not involve ρ ,

$$N\rho_1^* - \eta = \eta(N\alpha_D - 1) + N(1 - \alpha_D)(1 - \alpha_T).$$

At $\alpha_D = 1/N$,

$$N\rho_1^* - \sigma = (N-1)(1 - \alpha_T), \quad D_N = (N-1)\sigma,$$

so by (32),

$$\frac{d\log e_{AC}}{d\tau} = -\frac{1 - \alpha_T}{N\sigma}.$$

Relative treatment across partners. By (26), $\nu_j - \nu_k = \lambda_D(t_j - t_k)$ with $\lambda_D < 0$, so $t_j > t_k$ implies $\nu_j < \nu_k$. With $\bar{t} > 0$, A depreciates against partner j if and only if

$$\frac{\rho}{N}(\bar{t} - t_j) > \rho_1^* \bar{t} \quad \Longleftrightarrow \quad \frac{t_j}{\bar{t}} < 1 - \frac{N\rho_1^*}{\rho}. \quad (51)$$

To see this, note that $\nu_j > 0$ if and only if $\lambda_S \bar{t} + \lambda_D(t_j - \bar{t}) > 0$. Substituting (25) and dividing by $(N-1)/D_N > 0$ gives $-\rho_1^* \bar{t} - \frac{\rho}{N}(t_j - \bar{t}) > 0$, which rearranges to (51). The right-hand side of the second form is positive only if $\rho > N\rho_1^*$, so depreciation against a relatively lightly taxed partner requires both $\rho > N\rho_1^*$ and a sufficiently large tariff differential.

Tariffs on several partners. If A raises its tariff by τ on a set of k partners, $1 \leq k \leq N-1$, then for every untargeted partner C and every targeted partner B ,

$$\frac{d \log e_{AC}}{\tau} = \frac{k(\rho - N\rho_1^*)}{ND_N}, \quad \frac{d \log e_{AB}}{\tau} = -\frac{k\rho_1^*}{D_N} - \frac{(N-1-k)\rho}{ND_N}, \quad \frac{d\nu_A^E}{\tau} = -\frac{k\rho_1^*}{D_N}. \quad (52)$$

For every $k \leq N-2$, the bystander response has the same threshold $\rho > N\rho_1^*$. When all partners face the same tariff,

$$\frac{d \log e_{Aj}}{d\tau} = -\frac{(N-1)\rho_1^*}{D_N} = \frac{d\nu_A^E}{d\tau} \quad \forall j. \quad (53)$$

All bilateral rates then move together, and substitution across foreign origins plays no role.

To derive these, let $t_j = \tau$ on the k targets and zero otherwise, so that

$$\bar{t} = \frac{k\tau}{N-1}, \quad \tilde{t}_B = \frac{N-1-k}{N-1}\tau$$

for a target, and

$$\tilde{t}_C = -\frac{k\tau}{N-1}$$

for a bystander. Substituting into (26) gives

$$\nu_C = \frac{k\tau}{N-1}(\lambda_S - \lambda_D) = \frac{k\tau}{ND_N}(\rho - N\rho_1^*),$$

together with the target and NEER responses in (52). For every $k \leq N-2$, the sign of the bystander response is therefore determined by $\rho - N\rho_1^*$. At $k = N-1$ there is no bystander and $\tilde{t} = 0$, giving (53).

B.1 Three-country illustration

At $N = 3$, (50) and (49) give the unilateral-tariff system explicitly:

$$J = -\frac{mD_3}{4} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad F = -\frac{m}{4} \begin{pmatrix} \rho + \rho_1^* \\ \rho_1^* - \rho \end{pmatrix}, \quad D_3 = 3\alpha_D(\eta - 1) + \rho + 1. \quad (54)$$

With $D_3 > 0$, (31)–(32) read

$$\frac{d \log e_{AB}}{d\tau} = -\frac{\rho + 3\rho_1^*}{3D_3} < 0, \quad \boxed{\frac{d \log e_{AC}}{d\tau} = \frac{\rho - 3\rho_1^*}{3D_3}}, \quad \frac{d \log e_{BC}}{d\tau} = \frac{2\rho}{3D_3} > 0. \quad (55)$$

Thus A appreciates against B , B depreciates against C , and A depreciates against C if and only if $\rho > 3\rho_1^*$. By (7), $d\nu_C/d\tau \propto F_B + 2F_C \propto \rho - 3\rho_1^*$, so the sign of C 's own disturbance F_C , which is positive once $\rho > \rho_1^*$, does not settle the sign of ν_C .

Figure 3 plots the two equilibrium conditions,

$$\text{TB}_B = 0: \quad \nu_C = 2\nu_B + \frac{\rho + \rho_1^*}{D_3}d\tau, \quad \text{TB}_C = 0: \quad \nu_C = \frac{1}{2}\nu_B + \frac{\rho - \rho_1^*}{2D_3}d\tau. \quad (56)$$

In the intermediate case $\rho_1^* < \rho < 3\rho_1^*$, C gains from trade diversion at unchanged exchange rates and the $\text{TB}_C = 0$ locus shifts up. But the larger shift of $\text{TB}_B = 0$ drives ν_B down, and the resulting movement along the upward-sloping $\text{TB}_C = 0$ locus more than offsets that shift: A still appreciates against C .

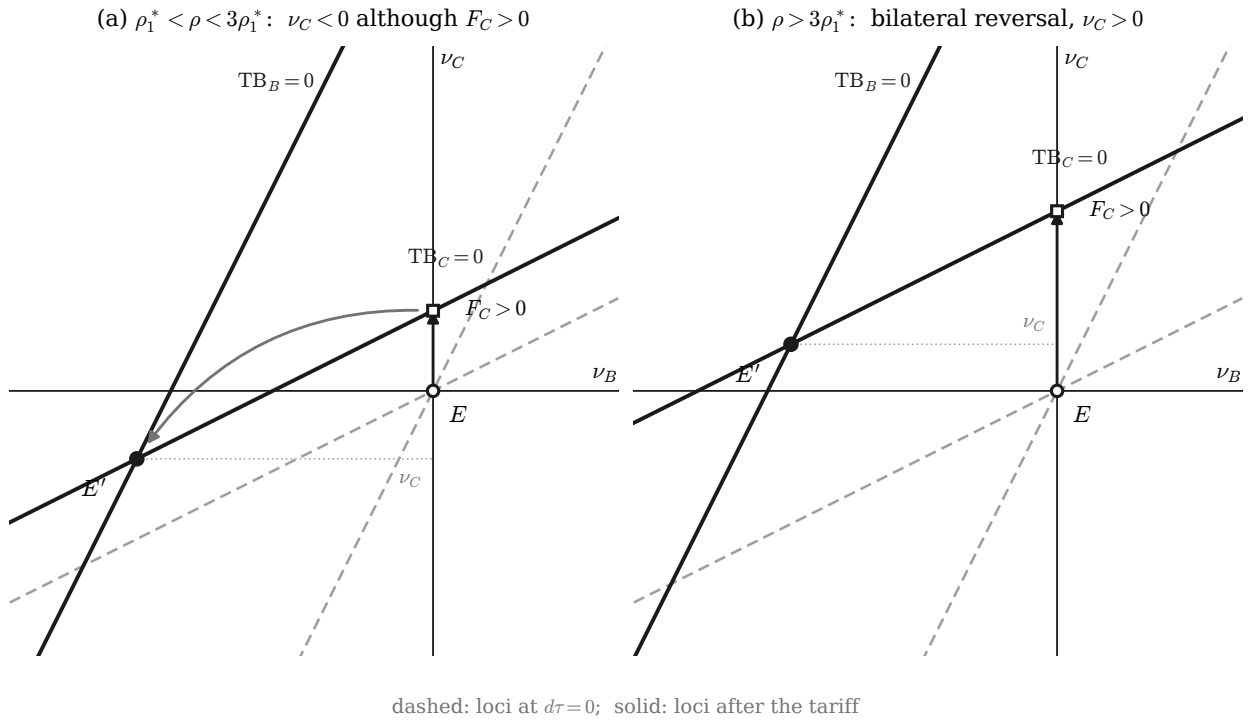


Figure 3: Three-country equilibrium adjustment. The loci are (56); dashed lines are the pre-tariff loci through the initial equilibrium E , solid lines the post-tariff loci intersecting at E' . The open square is the value of ν_C that would clear C 's trade balance at $\nu_B = 0$; the curved arrow in panel (a) is the subsequent movement along $\text{TB}_C = 0$. Panel (a): $\rho_1^* < \rho < 3\rho_1^*$. Panel (b): $\rho > 3\rho_1^*$. Drawn at $\alpha_D = 0.7$, $\alpha_T = 0.6$, $\eta = 1.5$, with $\rho = 2.2$ in panel (a) and $\rho = 5$ in panel (b), so that $\rho_1^* = 1.17$ and $3\rho_1^* = 3.51$.

C Real exchange rates

This appendix gives the real exchange-rate counterparts of the results in Sections 3.3–3.5. Real effective exchange-rate changes use the weights in (20), $dq_i^E \equiv \sum_{j \neq i} w_{ij} d \log q_{ij}$.

From (19) and (15),

$$d \log q_{Aj} = \nu_j + \alpha_T (d \log P_{Tj}^{\text{own}} - d \log P_{TA}^{\text{own}}).$$

For a tariff vector t , symmetry implies

$$d \log P_{MA} = d\nu_A^E + \bar{t}, \quad d \log P_{Mj} = d\nu_A^E - \frac{N}{N-1} \nu_j$$

for a partner that imposes no tariff. Hence

$$\begin{aligned} d \log q_{Aj} &= \left(1 - \frac{mN}{N-1}\right) \nu_j - m \bar{t}, \\ dq_A^E &= \left[\left(1 - \frac{mN}{N-1}\right) \lambda_S - m\right] \bar{t}, \\ d \log q_{Aj} - dq_A^E &= \left(1 - \frac{mN}{N-1}\right) \lambda_D \tilde{t}_j. \end{aligned} \tag{57}$$

Thus average protection determines the real effective exchange-rate response, while relative treatment determines bilateral deviations around it, subject to the price-index adjustment in (57).

For a unilateral tariff, $\bar{t} = d\tau/(N-1)$ and

$$d \log q_{Aj} = \left(1 - \frac{mN}{N-1}\right) \nu_j - \frac{m}{N-1} d\tau, \quad j = B, C. \tag{58}$$

Setting $d \log q_{AC} = 0$ and using (32), for $m < 1/N$ the real bilateral reversal threshold is

$$\rho_q^*(N) = \frac{[(N-1) - mN] N \rho_1^* + mN [N \alpha_D (\eta - 1) + 1]}{(N-1)(1 - mN)}. \tag{59}$$

Averaging (58) gives

$$\frac{dq_A^E}{d\tau} = -\left(1 - \frac{mN}{N-1}\right) \frac{\rho_1^*}{D_N} - \frac{m}{N-1} < 0 \quad \text{for all } \rho, N \text{ } (m < \frac{N-1}{N}). \tag{60}$$

Proposition C.1 (Real exchange rates). *At the symmetric benchmark of the nested-CES model, consider a unilateral tariff.*

- (i) *If $m < 1/N$ and MML holds at $\rho = \rho_q^*(N)$, the real bilateral reversal threshold exceeds the nominal threshold, $\rho_q^*(N) > N \rho_1^*$.*
- (ii) *If MML holds and $m < (N-1)/N$, the REER appreciates, $dq_A^E/d\tau < 0$.*

Both results reflect the direct price-index effect of the tariff. At a given nominal exchange rate, the tariff raises A 's consumer prices, making a real bilateral depreciation harder to obtain and

reinforcing the effective appreciation.

D Asymmetry

The disturbance-adjustment representation (5) and MML (6) do not require symmetry, and at any balanced baseline the linearized system (42)–(47) continues to hold with baseline shares and incomes as coefficients: b_{ij} , the share of origin j in i 's import bundle; h_i , the home share of i 's tradable expenditure; and $y_i \equiv \tilde{I}_i$, i 's income in the common currency. What requires symmetry is the collapse of the response to the two coefficients λ_S and λ_D of Proposition 2.3. Away from symmetry, G and $-J$ generally no longer act separately on average protection and relative treatment, so changes in relative treatment can affect the effective exchange rate. The unconditional NEER appreciation and the linear thresholds of Section 3 therefore need not survive. This appendix records what does.

Bilateral thresholds. The elasticity ρ enters the share responses (45) only through $(1 - \rho)$, so every entry of J and F is linear in ρ . Each entry of $\text{adj}(-J)$ is therefore a polynomial of degree at most $N - 2$ in ρ , and each numerator $[\text{adj}(-J)F]_i$ in (5) a polynomial of degree at most $N - 1$. Since $\det(-J) > 0$ under MML, a bilateral response can change sign only at a root of this numerator. Symmetry reduces the problem to the two coefficients of Proposition 2.3, yielding the linear thresholds (34) and (55). At a general asymmetric baseline the $N = 3$ numerator is already quadratic in ρ :

$$\text{sign } \frac{d \log e_{AC}}{d\tau} = \text{sign } (c_2 \rho^2 + c_1 \rho + c_0), \quad c_2 = b_{AB} b_{AC} b_{CA} b_{CB} (1 - h_A)(1 - h_C) y_A y_C. \quad (61)$$

The coefficients c_1 and c_0 depend on the full pattern of trade shares and country sizes. Provided MML continues to hold, $c_2 > 0$ whenever A and C each import from both foreign origins, so sufficiently strong substitution across import origins still reverses e_{AC} . When positive roots exist and MML continues to hold, the response is positive beyond the largest positive root, which equals $3\rho_1^*$ at the symmetric point. Away from symmetry there may be no positive root or several, so there is no universal bilateral threshold, and for $N \geq 4$ a response may change sign more than once as ρ varies.

E Correspondence across N

All entries are at the symmetric benchmark of the nested-CES model.

object	$N = 2$	$N = 3$	general N
adjustment condition	$D_2 = \varepsilon_X + \varepsilon_M - 1 > 0$	$D_3 > 0$	$D_N > 0$ (MML)
tariff-disturbance coefficients g_S, g_D (27)	$-m\rho_1^*, -$	$-m\rho_1^*/2, -m\rho/2$	$-m\rho_1^*/(N-1), -m\rho/(N-1)$
adjustment coefficients j_S, j_D (28)	$mD_2, -$	$mD_3/4, 3mD_3/4$	$mD_N/(N-1)^2, NmD_N/(N-1)^2$
disturbance and adjustment ratios $g_D/g_S, j_D/j_S$	$-$	$\rho/\rho_1^*, 3$	$\rho/\rho_1^*, N$
unilateral bystander threshold	none	$3\rho_1^*$	$N\rho_1^*$
unilateral NEER response	$-\rho_1^*/D_2$	$-\rho_1^*/D_3$	$-\rho_1^*/D_N$
trade-war threshold	$-$	ρ_1^*	ρ_1^*

F Proofs

Throughout, $m \equiv \alpha_T(1 - \alpha_D)$ and all derivatives are evaluated at the balanced free-trade baseline of Appendix B.

Proof of Lemma 2.2. Scaling every currency value e_{Al} by a common factor leaves all relative prices in (42) unchanged and scales every common-currency income, and hence every flow f_{ij} and trade balance TB_i , by the same factor. Thus TB_i is homogeneous of degree one in the currency values. By Euler's theorem, $\sum_l \tilde{J}_{il} = TB_i$, which is zero at the balanced baseline, so $\sum_l \tilde{J}_{il} = 0$ for all i . Equivalently, in (47), a common shift in all ν_l affects only the income terms, whose export and import contributions cancel by baseline balance. Summing (18) over i gives $\sum_i TB_i \equiv 0$, so $\sum_i \tilde{J}_{il} = 0$ for all l . Under Assumption 1, $\tilde{J}_{il} \geq 0$ for $l \neq i$, so $\tilde{J}_{ii} = -\sum_{l \neq i} \tilde{J}_{il} \leq 0$. Deleting row and column A then gives

$$|J_{ii}| - \sum_{l \neq i, A} J_{il} = \tilde{J}_{iA} \geq 0, \quad (62)$$

with strict inequality in every row for which $\tilde{J}_{iA} > 0$.

Thus $-J$ is a Z-matrix with nonpositive off-diagonal entries, nonnegative diagonal entries, weak diagonal dominance, irreducibility, and strict diagonal dominance in at least one row. Its diagonal is strictly positive, since $|J_{ii}| = 0$ would force row i of J to vanish, contradicting irreducibility. Write $-J = D - B$, where $D = \text{diag}(-J) > 0$ and $B \geq 0$. Diagonal dominance implies that every row sum of $D^{-1}B$ is at most one and that at least one is strictly below one. With irreducibility, its spectral radius therefore satisfies $r \equiv \rho(D^{-1}B) < 1$. Hence

$$(-J)^{-1} = (I - D^{-1}B)^{-1}D^{-1} = \sum_{k \geq 0} (D^{-1}B)^k D^{-1} \geq 0.$$

Nonsingularity follows from $r < 1$. Moreover,

$$\det(-J) = \det D \cdot \det(I - D^{-1}B) > 0.$$

□

Proof of Proposition 2.3. Let M be an $(N-1) \times (N-1)$ matrix with $PMP' = M$ for every permutation matrix P . Conjugating by the transposition of i and j gives $M_{ii} = M_{jj}$, and conjugating by a permutation that sends the ordered pair (i, j) to (k, l) , $i \neq j$, $k \neq l$, gives $M_{ij} = M_{kl}$. Hence

$$M = aI + b\mathbf{1}\mathbf{1}'.$$

Since $\mathbf{1}'\mathbf{1} = N-1$,

$$M\mathbf{1} = [a + (N-1)b]\mathbf{1},$$

while $Mv = av$ for every v satisfying $\mathbf{1}'v = 0$. This gives (14) for $-J$ and G .

For $j_S, j_D \neq 0$,

$$(-J)^{-1} = \frac{1}{j_D}I + \frac{1}{N-1}\left(\frac{1}{j_S} - \frac{1}{j_D}\right)\mathbf{1}\mathbf{1}',$$

as direct multiplication verifies. Thus $(-J)^{-1}$ and G share the constant and zero-sum eigenspaces, with response coefficients g_S/j_S and g_D/j_D . Writing

$$t = \bar{t}\mathbf{1} + \tilde{t}, \quad \mathbf{1}'\tilde{t} = 0,$$

gives

$$\nu = \frac{g_S}{j_S}\bar{t}\mathbf{1} + \frac{g_D}{j_D}\tilde{t},$$

and averaging with equal weights yields

$$d\nu_A^E = \frac{g_S}{j_S}\bar{t}.$$

For (13), the off-diagonal entry of $-J$ is

$$b_J = \frac{j_S - j_D}{N-1},$$

so $-J$ has the Z-matrix sign pattern if and only if $j_D \geq j_S$. Given $j_D \geq j_S > 0$, the entries of $(-J)^{-1}$ above are nonnegative, so MML holds. Conversely, if $j_S \leq 0$, either $-J$ is singular or its inverse cannot be elementwise nonnegative. At $N = 2$ the zero-sum subspace is trivial and the condition reduces to $j_S > 0$.

Finally, for any two partners $j, k \neq A$,

$$d \log e_{jk} = \nu_k - \nu_j = \lambda_D(\tilde{t}_k - \tilde{t}_j) = \lambda_D(t_k - t_j),$$

which completes (12). □

Proof of Theorem 3.1. At the symmetric benchmark relabeling A 's partners leaves the reduced system unchanged, so Proposition 2.3 applies. Appendix B derives (48), (27) and (28). By (28), $j_S = mD_N/(N-1)^2$ and $j_D = Nj_S$, so $j_D \geq j_S$ always and $j_S > 0$ if and only if $D_N > 0$; by (13), MML is therefore equivalent to $D_N > 0$. If $\eta \geq 1$ then every term of (24) is nonnegative and the last is one, so $D_N > 0$. Given $D_N > 0$, dividing (27) by (28) gives (25),

$$\lambda_S = \frac{g_S}{j_S} = -\frac{m\rho_1^*}{N-1} \cdot \frac{(N-1)^2}{mD_N} = -\frac{(N-1)\rho_1^*}{D_N}, \quad \lambda_D = \frac{g_D}{j_D} = -\frac{(N-1)\rho}{ND_N},$$

both negative since $\rho_1^* > 0$ by (21) and $\rho \geq 0$. Equation (26) and the three statements following it are then (11) and (12). $\square$

Proof of Proposition 4.1(i). Let primes denote responses to the mirror shock $t_{BA} = d\tau$. Relabeling $A \leftrightarrow B$ in (31) gives

$$d \log e'_{BA} = -\frac{N\rho_1^* + (N-2)\rho}{ND_N}, \quad d \log e'_{BC} = \frac{\rho - N\rho_1^*}{ND_N}.$$

Since $e_{AB} = 1/e_{BA}$ and $e_{AC} = e_{AB}e_{BC}$,

$$d \log e'_{AB} = \frac{N\rho_1^* + (N-2)\rho}{ND_N}, \quad d \log e'_{AC} = \frac{(N-1)\rho}{ND_N}.$$

By linearity,

$$d \log e_{AB}|^w = 0,$$

and

$$d \log e_{AC}|^w = \frac{\rho - N\rho_1^*}{ND_N} + \frac{(N-1)\rho}{ND_N} = \frac{\rho - \rho_1^*}{D_N}.$$

In terms of Theorem 3.1, the three components are A 's average protection, A 's relative treatment, and B 's relative treatment:

$$d \log e_{AC}|^w = -\frac{\rho_1^*}{D_N} + \frac{\rho}{ND_N} + \frac{(N-1)\rho}{ND_N} = \frac{\rho - \rho_1^*}{D_N}.$$

By symmetry between A and B , the same expression applies to $d \log e_{BC}|^w$. Averaging gives

$$\left. \frac{d\nu_A^E}{d\tau} \right|^w = \frac{N-2}{N-1} \frac{\rho - \rho_1^*}{D_N}.$$

Under MML, $D_N > 0$, so the signs are determined by $\rho - \rho_1^*$. $\square$

Proof of Proposition 4.1(ii). Superpose the N tariff vectors. A 's tariff vector is uniform,

$$\bar{t}^{(A)} = d\tau, \quad \tilde{t}^{(A)} = 0,$$

and contributes $\lambda_S d\tau$ to every $d \log e_{Aj}$.

Partner j taxes only A . Its contribution to $d \log e_{Aj}$ is

$$-[\lambda_S + (N-2)\lambda_D] \frac{d\tau}{N-1},$$

while each of the $N-2$ other partners contributes $-\lambda_D d\tau$. Summing,

$$\begin{aligned} d \log e_{Aj} &= \lambda_S d\tau - [\lambda_S + (N-2)\lambda_D] \frac{d\tau}{N-1} - (N-2)\lambda_D d\tau \\ &= \frac{(N-2)(\rho - \rho_1^*)}{D_N} d\tau. \end{aligned}$$

The expression is the same for every j , so it also equals the NEER response, and all cross rates among A 's partners are unchanged. $\square$

Proof of Proposition C.1. From (19) and (15), only tradable prices move, so

$$d \log q_{Aj} = \nu_j + \alpha_T (d \log P_{T_j}^{\text{own}} - d \log P_{T_A}^{\text{own}}),$$

with

$$d \log P_{T_i}^{\text{own}} = (1 - \alpha_D) d \log P_{M_i}.$$

For a tariff vector t ,

$$d \log P_{M_A} = \frac{1}{N-1} \sum_j (\nu_j + t_j) = d\nu_A^E + \bar{t},$$

while for a partner that imposes no tariff,

$$d \log P_{M_j} = d\nu_A^E - \frac{N}{N-1} \nu_j.$$

Substitution gives (57), and $\bar{t} = d\tau/(N-1)$ gives (58).

For (i), setting $d \log q_{AC} = 0$ and substituting ν_C from (32) gives

$$[(N-1) - mN] \frac{\rho - N\rho_1^*}{N} = mD_N(\rho).$$

For $m < 1/N$, solving for ρ gives (59). The coefficient $(N-1) - mN$ on the left is then positive, and MML at $\rho = \rho_q^*(N)$ gives $D_N > 0$ on the right, so $\rho_q^*(N) > N\rho_1^*$. For (ii), averaging (58) and using (33) gives (60); under MML $D_N > 0$, and for $m < (N-1)/N$ both terms are negative. $\square$